Case dossier #F1BCFEBD

Awaiting review

Tuberous sclerosis complex

Opened Jul 31, 2026, 03:34 AM Run cost $0.025

Pipeline complete — awaiting your sign-off

Repurposing hypothesis: EBASTINE → Tuberous sclerosis complex

NOTE: This candidate did NOT meet the STRONG_MATCH threshold (composite 0.5696 < 0.70). It is included as the highest-ranked hypothesis for review; treat it accordingly.

Repurposing-only pool: the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max_phase ≥ 4) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

ℹ Top-K evaluation: 5 of 5 target(s) successfully evaluated.

ℹ **Pathway-neighbor candidate.** The target **PRKAA2** was not directly linked to **Tuberous sclerosis complex** via Open Targets; it was discovered because it co-participates in the same Reactome pathway(s) as the primary causal gene. Open Targets association score for this target: 0.000 (0 = no direct link). The drug–target binding evidence (pChEMBL, confidence) is real; only the disease-relevance link is inferred from pathway adjacency.

⚠ **High lipophilicity (XLogP = 7.20) — empirical caution flag.** **EBASTINE** has a PubChem XLogP of 7.20, above the threshold of 5. A dataset-derived analysis (repoDB, n ≈ 8,700 drug–disease pairs) found that drugs with XLogP ≥ 5 have **0.426× the odds of repurposing success** (Fisher's exact $p = 3\times10^{-9}$; holdout p = 0.009; survives adjustment for established-product status and CNS-area membership). The direction of effect is opposite to the original hypothesis — high-lipophilicity drugs repurpose *less* often, not more. **Unresolved caveat:** an incumbency-confound (lipophilic drugs are disproportionately represented in historically studied disease areas) has not yet been fully ruled out; treat this as an alert rather than a disqualifier. **This flag does not affect the composite score.** The reviewer must judge whether lipophilicity is a material concern in the context of the proposed repurposing indication.

1. Hypothesis summary

EBASTINE is proposed as a repurposing candidate against **Tuberous sclerosis complex** via the target **PRKAA2**. It shows a ChEMBL median pChEMBL affinity of 7.58 at assay confidence 9/9, an Open Targets target-disease association of 0.000, and a Tanimoto similarity of 0.155 to ENTRECTINIB. Target network context (BioGRID, physical/genetic — not mechanism): TGFBR1, YY1, MTOR, RPTOR, DHFR, RYR1, RYR3, ITPR1. The resulting composite score is 0.5696.

Chemist rationale: Ebastine has a median pChEMBL affinity of 7.58 against PRKAA2 at an assay confidence score of 9 out of 9, and it is an approved or known drug with a Tanimoto similarity of 0.155 to the nearest approved drug entrectinib. The PRKAA2 target has eight recorded BioGRID interactors—MSN, YWHAZ, EZR, ACTA1, AKT1, NEFL, RHEB, and RPS6KA1.

STAGE 1 PRIORITIZATION SCORES

- **tractability_score:** 0.6632 (ChEMBL bioactivity + AlphaFold pLDDT − prior-trial-failure penalty)
- **unmet_need_score:** 0.0410 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

2. Evidence table

Evidence	Value
ChEMBL median pChEMBL affinity	7.58
Assay confidence score (0-9)	9
Open Targets association score	0.000
Tanimoto to nearest approved drug	0.155 (ENTRECTINIB)
Approved / known drug	yes
Mutation-specific approved indication (disclosure)	No specific mutation named in approved indication
Lipinski/Veber (MW, logP, HBD, HBA, TPSA, rotB)	469.7, 7.22, 0, 3, 29.5, 9
Lipinski violations / Veber pass	1 / yes
PubChem XLogP (lipophilicity)	⚠ 7.20 (≥ 5 — empirical caution flag; see high-lipophilicity disclosure above)

Evidence	Value
AFDB apo structure mean pLDDT (free protein, no ligand)	76.7
Boltz structure confidence (0-1)	0.724
Boltz binding-pose confidence (0-1)	0.153
Boltz predicted affinity (relative optimization score, 0-1, NOT a Kd)	0.240
Boltz predicted structure (CIF)	Download CIF (/api/structures/e9309a1002f3befb170a5998446ebd00.cif)
Boltz ADME — lipophilicity (logD)	4.933
Boltz ADME — permeability	0.557
Boltz ADME — solubility	high-risk
openFDA adverse-event signal (FAERS)	URTICARIA (342), PRURITUS (301), FATIGUE (270), COVID-19 (263), NAUSEA (236)
Prior trials for this exact drug+disease	0
Target discovery method	pathway_neighbor

3. Full source citations

- **PMIDs (0):** none
- **ChEMBL activity IDs (1):** 25872880
- **NCT numbers (0):** none

4. Composite score breakdown

Term	Weight	Component value	Contribution
Normalized pChEMBL affinity	0.30	0.654	0.1963
Assay confidence (score / 9)	0.20	1.000	0.2000
Normalized Open Targets association	0.20	0.000	0.0000
Normalized Tanimoto similarity	0.15	0.155	0.0233
No prior failed trial (1/0)	0.15	1	0.1500
Composite (weighted sum - penalty - cap)			**0.5696**

Weighted sum before penalty = 0.5696; penalty = 0.0000; reported composite_score = 0.5696.

5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.153) or predicted affinity (0.240) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.
- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure_confidence of 0.724 (complex pLDDT 0.686) and an AFDB apo mean pLDDT of 76.7; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.

- **Assay-type and species caveats.** The affinity is a median pChEMBL over Homo sapiens IC50/Ki assays at confidence ≥ 8; assay heterogeneity and the bounded approved-drug reference set for Tanimoto still apply.
- **Absence of evidence is not evidence of absence.** A zero prior-trial count or no adverse-event signal may reflect that the pair has simply never been studied, not that it is safe or untried.
- **This is a repurposing *hypothesis*, not a finding.** It is a prioritised starting point that requires wet-lab and, ultimately, clinical validation.

TARGET DRUGGABILITY CONTEXT

- **Approved drugs with known mechanism against this target (ChEMBL):** 5 — EVEROLIMUS, PIMECROLIMUS, SIROLIMUS, TACROLIMUS, TEMSIROLIMUS
- **Historical difficulty literature:** insufficient signal found (fewer than 2 qualifying abstracts in targeted PubMed searches for undruggability / resistance / difficulty).

Druggability context is informational only. It does not affect tractability_score, unmet_need_score, composite_score, or STRONG_MATCH.

HUMAN REVIEW CHECKPOINT

This is a falsifiable hypothesis, not a finding.

Record your scientific reasoning below, then sign off. Approving advances this candidate for wet-lab validation; rejecting closes the case permanently.

REVIEWER NOTE required

e.g. Affinity and structure confidence justify wet-lab follow-up despite the sub-threshold composite score.

Approve case

Reject case